KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Electronics and Communication Engineering

PROPOSAL

Analog Electronics – II Laboratory
Course Code: ECE – 2102

PROJECT TITLE

Design and Implementation of an IR-Based Pulse Detector Using Operational Amplifiers

Submitted by,

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1. Background and Motivation

Heart rate is an important physiological parameter that provides useful information about cardiovascular activity. Monitoring pulse rate is therefore valuable for health awareness and understanding basic biomedical instrumentation. This project focuses on developing a low-cost, analog electronic pulse detector using infrared (IR) technology and operational amplifiers.

The proposed system employs an IR light-emitting diode to illuminate the fingertip and a photodiode to detect variations in received light associated with pulsatile changes in blood volume. These variations generate a weak electrical signal that requires amplification and conditioning for reliable detection. Two LM358 operational amplifier stages are used to process the sensor signal and identify pulse-related changes, while an LED provides a visual indication of detected pulses. The circuit operates from a single 5 V DC supply, making it suitable for a simple, low-power prototype.

The motivation behind this project is to apply fundamental concepts of analog electronics, including optical sensing, voltage amplification, biasing, and signal conditioning, to a practical application. By using readily available and inexpensive components, the project demonstrates the feasibility of constructing a basic pulse-detection system while developing practical skills in circuit design, implementation, and troubleshooting. The prototype is intended for educational purposes rather than medical diagnosis.

2. Objectives

1. To design and implement an IR-based pulse detector using a photodiode and LM358 operational amplifiers.
2. To detect pulse-related variations in the received infrared light caused by changes in fingertip blood volume.
3. To amplify and condition the weak sensor signal using appropriate analog circuit techniques.
4. To develop an LED-based visual indication system for detecting pulse events.
5. To investigate the performance, sensitivity, and stability of the implemented circuit.
6. To gain practical experience in analog circuit design, component selection, hardware implementation, testing, and troubleshooting.

3. Conflicting Requirements

1. **Sensitivity vs. Position Tolerance:** The detector responds to pulse signals only when the fingertip is positioned within a suitable region, making consistent detection difficult when finger placement varies.
2. **Sensitivity vs. Stability:** The circuit must detect weak pulse-related signals while maintaining stable LED blinking without false triggering or continuous illumination.
3. **Signal Amplification vs. Noise:** Sufficient amplification is required to detect small optical variations, but excessive gain may amplify unwanted noise and interference.
4. **Detection Reliability vs. Circuit Simplicity:** Achieving consistent pulse detection requires careful sensor alignment and circuit adjustment, while the design aims to remain simple and inexpensive.
5. **Detection Performance vs. Power Consumption:** The IR LED must provide adequate illumination for reliable sensing while operating within safe current limits and minimizing power consumption.

4. Proposed Budget for Implementing the Project

Table-01: Table for the proposed expenses for this project

Sl.	Item	Justification	Price (BDT)
1	IR LED and BPV10NF photodiode	Infrared transmission and pulse sensing	20
2	2 × LM358P ICs	Signal amplification and pulse detection + 1 backup IC	40
3	Resistors and capacitors	Biasing, amplification, and signal conditioning	30
4	Breadboard and Wiring Kit	Circuit assembly and interconnection	300
5	LED and miscellaneous components	Visual indication and supporting connections	20
Total (BDT): 410 BDT (approx.)			
In words: Four hundred and ten taka only.			

5. Work Plan (Week-wise Plan)

1st Week: Project Planning and Circuit Design

- Study the operating principle of infrared-based pulse detection.
- Select the required components and determine their specifications.
- Design the sensor, amplifier, and pulse-detection stages.
- Perform preliminary circuit analysis and simulation.

2nd Week: Hardware Implementation and Testing

- Assemble the circuit using the IR LED, photodiode, LM358 operational amplifiers, and supporting components.
- Test the amplification and pulse-detection stages using a regulated 5 V supply.
- Troubleshoot circuit connections and adjust component values to obtain pulse-responsive LED operation.

3rd Week: Performance Optimization and Documentation

- Optimize sensor positioning and circuit sensitivity for more stable pulse detection.
- Observe and record the output response under different fingertip positions.
- Verify circuit operation and collect relevant experimental measurements.
- Analyze the results, document the limitations, and prepare the final project report.

6. Impact of Project on Society, Environment and Sustainability

Societal Impact: The project demonstrates an affordable method of detecting pulse-related changes using infrared technology and basic electronic components. It promotes awareness of optical pulse monitoring and provides practical knowledge of biomedical instrumentation. The prototype can serve as an educational tool for students studying analog electronics, sensor technology, and biomedical applications. Furthermore, the project illustrates how fundamental electronic principles can be applied to develop simple health-monitoring prototypes at a relatively low cost. However, the system is intended for educational purposes and is not suitable for medical diagnosis or clinical monitoring.

Environmental Impact: The circuit operates from a low-voltage DC supply and uses a relatively small number of electronic components, resulting in modest resource requirements during operation. Reusing breadboards, connecting wires, and laboratory equipment can reduce material consumption and electronic waste. The low-power design also helps limit energy consumption. Proper maintenance, component reuse, and responsible disposal of damaged electronic parts can further minimize the environmental impact.

Sustainability: The use of inexpensive and readily available components makes the project economical to maintain, repair, and modify. Individual components can be replaced when necessary, extending the useful life of the prototype and reducing the need to discard the entire circuit. The design can also serve as a foundation for future improvements, including better sensor positioning, more stable pulse detection, and automatic pulse-rate calculation. These possibilities support continued technical learning, affordable prototyping, and the sustainable development of basic electronic instrumentation.

Signature and Name of the Supervisor